

SPECTROSCOPY

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Spectroscopy is the study of interaction of electromagnetic radiation with matter. Ordinary light is an electromagnetic radiation, travelling in space in the form of waves. It consists of oscillating electric and magnetic fields in mutually perpendicular directions and has the following characteristics.

1. Wave length λ – distance travelled during a complete cycle, unit-meters.
2. Frequency ν is the number of waves produced in one second, unit- Hertz(Hz) $1\text{Hz} = \text{s}^{-1}$
3. Frequency $\nu = c / \lambda \text{ s}^{-1}$
4. Velocity of electromagnetic waves are similar to velocity of light $c = 3 \times 10^8 \text{ ms}^{-1}$.
5. Energy $E = h \nu = hc / \lambda$, or $E/hc = 1/\lambda = \bar{\nu} \text{ cm}^{-1}$

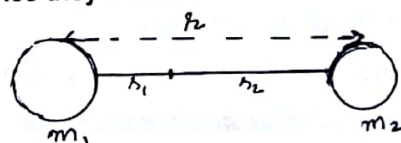
When light radiation falls on molecules some electric or magnetic effect present in molecular motion can be influenced by the electric or magnetic fields associated with the radiation. Different Possibilities are discussed below:

SPECTRA	TRANSITIONS IN MOLECULE	REGION OF SPECTRA	CRITERIA
1.MICROWAVE(ROTATIONAL)	Between rotational energy levels	Microwave (1 to 100cm^{-1})	Molecule must possess permanent dipole moment
2.VIBRATIONAL(INFRA-RED)	Between vibrational energy levels	infra-red (500 to 4500cm^{-1})	Dipole moment must change during vibration
3.ELECTRONIC (UV-VIS)	Between electronic energy levels	UV (100 to 400nm) Visible(400 to 800nm)	Presence of chromophoric group in a molecule.
4.NMR	Between magnetic energy levels of nuclei	Radiofrequency region	Half α -integral spin of nucleus

MICROWAVE OR ROTATIONAL SPECTROSCOPY

Certain molecules are dipolar. One end of the molecule carries permanent +ve charge and the other end has permanent -ve charge. Such molecules have permanent fluctuating electric dipole moment .Hence interaction can occur with electric component of electromagnetic radiation. Absorption of microwave radiation causes the rotation of the dipole moment of the molecule around various axes. Thus all molecules having a permanent dipole moment are said to be microwave active. HF, HCl, HBr, HI, NH_3 , H_2O , CH_3Cl etc. are some microwave active molecules. H_2 , Cl_2 etc have no dipoles in molecules , so no interaction take place, hence they are microwave inactive.

Consider a diatomic molecule in which m_1 and m_2 are the masses of two atoms and 'r' is the equilibrium bond length, rotating about an axis passing through the centre of gravity.



$$r = r_1 + r_2$$

The angular momentum L of rotating molecule is quantized and is given by

$$L = \sqrt{J(J+1)} \frac{h}{2\pi}, \text{ rotational quantum number } J=0,1,2,3... \quad \text{---(1)}$$

$$\text{The energy of rotating molecule is given by } E_J = \frac{L^2}{2I} \quad \text{---(2)}$$

Where moment of inertia $I = \mu r^2$, where μ is the reduced mass of the molecule. $\mu = \frac{m_1 m_2}{m_1 + m_2}$

Substituting (1) in (2) and dividing by hc we get, $\frac{E_J}{hc} = \frac{h}{8\pi^2 I c} J(J+1) \text{ cm}^{-1} = \bar{\nu}_J \text{ cm}^{-1}$ — (3)

or $\bar{\nu}_J = B J(J+1)$, where $J=0,1,2,3,\dots$ — (4) (J is called the rotational quantum number)

Where ' B ' = $\frac{h}{8\pi^2 I c}$ is a constant called the rotational constant expressed in wave numbers

Thus rotational energy levels can be calculated by substituting various J values in (4) as shown below.

Selection rule for obtaining rotational spectra

is $\Delta J = \pm 1$

Selection rule states that the allowed rotational energy transitions are

$J = 0 \rightarrow 1, 1 \rightarrow 2, 2 \rightarrow 3$ etc. or the reverse.

Transition between $J = 1 \rightarrow 3, 2 \rightarrow 5$ etc are not allowed.

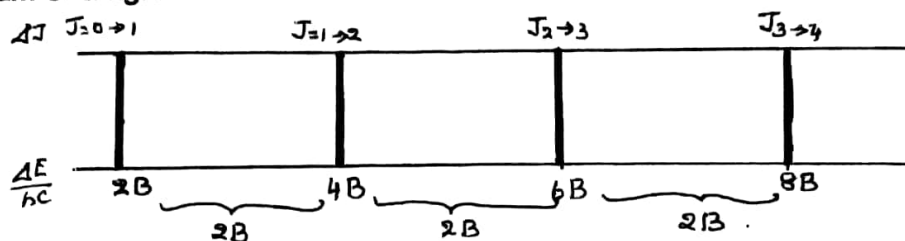
For a transition from J to $J+1$, the rotational frequency is given by

$$E_{J \text{ to } J+1} = 2B(J+1) \text{ cm}^{-1} \text{ — (5)}$$

Thus $E_{(J=0 \text{ to } J=1)} = 2B$, $E_{(J=1 \text{ to } J=2)} = 4B$,

$E_{(J=2 \text{ to } J=3)} = 8B$ etc.

Thus rotational spectrum of a rigid diatomic molecule consists of a series of equally spaced lines at $2B, 4B, 6B, 8B$ etc.



spacing between lines = $2B$ Rotational spectrum.

APPLICATIONS OF MICROWAVE SPECTROSCOPY

1. Determination of bond lengths in polyatomic molecules.
2. To study internal rotation of the molecule
3. It can readily distinguish the presence of isotopes in a sample.
4. Determination of symmetry in a molecule.
5. Determination of dipole moments of gaseous molecules.
6. To study the inversion spectrum of ammonia.
7. Microwave analysis is used in chemical examination of interstellar space (space between stars)

Problems:- refer class notes.

FTIR - SPECTROSCOPY

Molecules with permanent dipole moment or change in dipole moment during vibration exhibit IR spectra.

This change in dipole moment generates an oscillating electric field which resonates with the electric field of radiation and thus energy can be transferred from radiation to the molecule. CO, NO, HCl etc are IR active while H_2, N_2, O_2 etc are IR inactive (because they have no dipole moment).

Vibration of a diatomic molecule with masses m_1 and m_2 is similar to a simple harmonic oscillator (SHO).

$$\text{Vibrational frequency} = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}$$

where k is the force constant and $\mu = \frac{m_1 m_2}{m_1 + m_2}$

is the reduced mass

The quantized vibrational energy levels are given by

$$E_{vib} = (v + \frac{1}{2}) h \nu$$

$$\text{or } \frac{E_{vib}}{hc} = (v + \frac{1}{2}) \frac{h \nu}{c} = (v + \frac{1}{2}) \omega_e = \bar{\nu} \text{ cm}^{-1}$$

Where ω_e is the equilibrium vibrational frequency and v is the vibrational quantum number, $v=0, 1, 2, 3, \dots$

The lowest vibrational level when $v_0 = 0$ is $\bar{\nu} = \frac{1}{2} h \nu$

This is called zero point energy

Energy levels of SHO are equally spaced, spacing being equal to $h \nu$

The selection rule for a vibrational transition is

$$\Delta v = \pm 1$$

In actual practice, the molecule does not behave as SHO. Anharmonicity arises. The energy levels of anharmonic oscillators are given by

$$E_{vib} = (v + \frac{1}{2}) h \nu_0 - (v + \frac{1}{2})^2 h \nu_0 x_e \text{ joules}$$

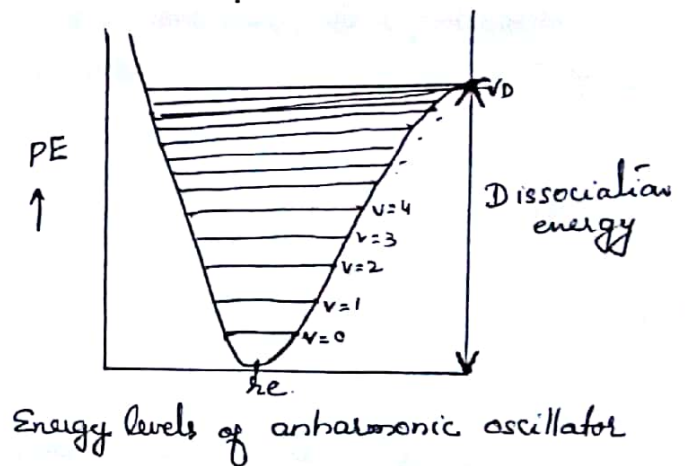
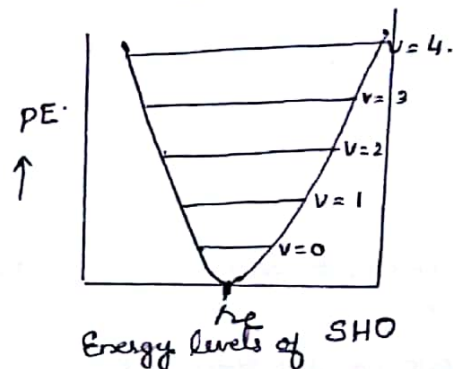
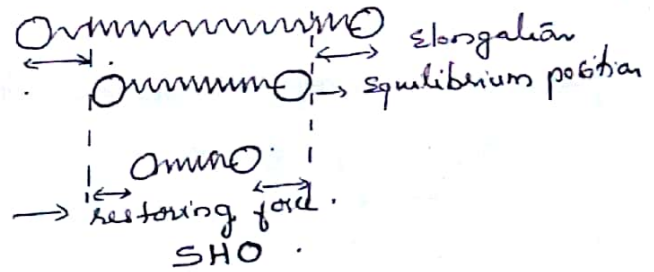
In this case the spacing between the vibrational levels goes on decreasing with increase in quantum number.

Selection rule for vibrational transition for anharmonic oscillator are

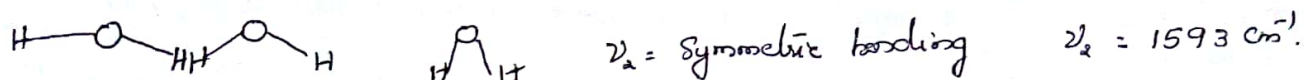
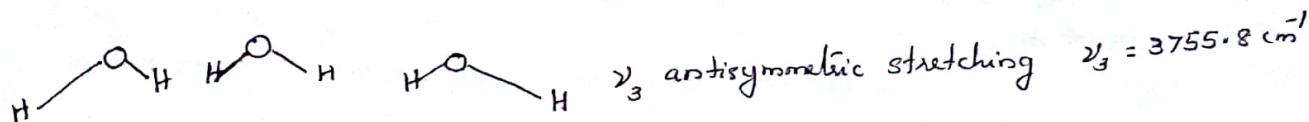
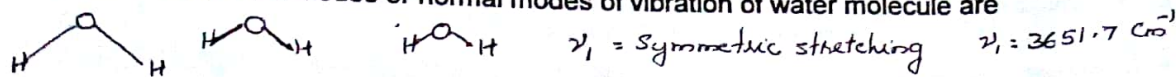
$$\Delta v = \pm 1, \pm 2, \pm 3, \dots$$

The three transitions from $v = 0$ are

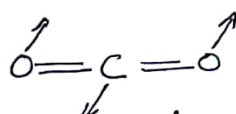
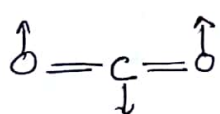
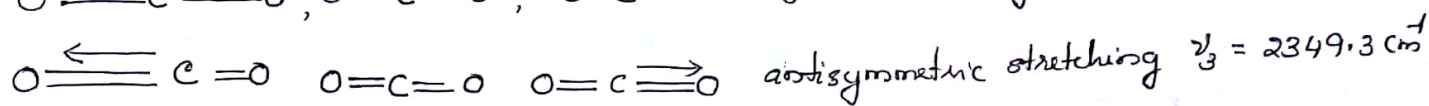
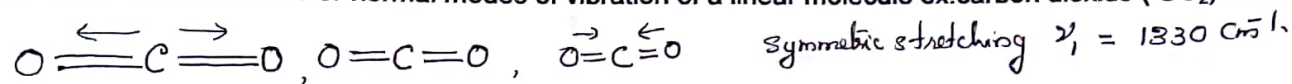
$v = 0 \rightarrow v = 1$ called Fundamental frequency, $v = 0 \rightarrow v = 2$ - first overtone, $v = 0 \rightarrow v = 3$ - second overtone. Besides these combination bands also occur. Thus IR spectrum of even a simple diatomic molecule may contain a great many lines. In the case of polyatomic molecules containing N atoms, the total number of degrees of freedom is $3N$, of which 3 are translational degrees and 3 are rotational. The remaining $3N - 6$ is the vibrational degrees of freedom. For a linear molecule it is $3N - 5$. These are called the fundamental modes or normal modes of vibration.



The fundamental modes or normal modes of vibration of water molecule are



The fundamental modes or normal modes of vibration of a linear molecule ex: carbon dioxide (CO_2) molecule are



Bending vibration (degenerate) $\nu_2 = 667.3 \text{ cm}^{-1}$

The normal modes may be divided into two classes 1. Fingerprint vibrations or skeletal bands. 2. Characteristic group vibrations.

Fingerprint bands or skeletal bands: this fall in the low frequency region 1500cm^{-1} to 700cm^{-1} . The matching of the finger print region with the spectrum of a known compound is a very powerful way of confirming the presence of a particular substance.

Characteristic group or functional group vibrations: This region lie between 1500cm^{-1} to 4000cm^{-1} is very useful for the identification of functional groups of unknown compounds. For example:

Groups	Region cm^{-1}	Groups	Region cm^{-1}
-OH	3650-3590	C=O	1780-1680
-C=C-	2500-2000		3060
C=C	1680-1620	-NH ₂	3400

APPLICATIONS OF IR- SPECTROSCOPY

Determination of force constant

Identification of unknown compounds

Identification of functional groups in organic molecules.

To distinguish between intra and inter molecular hydrogen bonding.

Determination of molecular structure.

Determination of purity.

Electronic spectra arise from electronic transitions in a molecule by the absorption of visible and ultraviolet radiations. Visible region is 400 to 800 nm and UV- region span from 200 to 400nm. According to

Born- Oppenheimer approximation

Electronic transitions in a molecule are always accompanied by vibrational and rotational transitions. Hence a broadened spectra will be obtained.

total energy $E = E_{el} + E_{vib} + E_{rot}$ (neglecting translational energy.)

Energy change for an electronic transition is given by $\nu = \frac{\Delta E}{hc}$

The intensity of spectra depends on path length and concentration of the sample. The relationship between the path length and concentration of the sample can be expressed on the basis of **Beer-Lambert Law**

Beer's Law: The rate of decrease in intensity of light when it passes through a solution is directly proportional to the concentration of the solution. $-dI = k'C \dots \dots \dots (1)$

Lambert's Law: The rate of decrease in intensity of light when it passes through a medium is directly proportional to the thickness dx and intensity 'I' of incident radiation. $-dI = k''I dx \dots \dots \dots (2)$

Combining the two relationships, we get $-dI = kCIdx \dots \dots \dots (3)$

Integrating within the limits $I=I_0$ to I and $x=0$ to l

$$\int_{I_0}^I \frac{dI}{I} = -kC \int_{x=0}^l dx$$

$$\ln \frac{I}{I_0} = 2.303 \log \frac{I}{I_0} = -kCl$$

$$\therefore \frac{I}{I_0} = e^{-kCl}$$

The intensity of a beam of monochromatic radiation decreases exponentially with increase in thickness 'l' and concentration C of the absorbing medium. This is Beer -Lambert law.

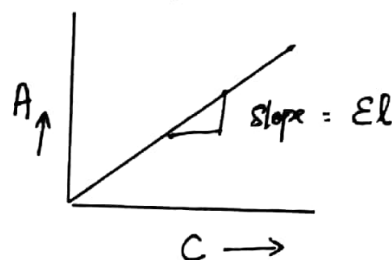
If $\log \frac{I_0}{I}$ is represented as absorbance A and $k/2.303$ as ϵ then $A = \epsilon Cl$

ϵ is called as molar absorptivity coefficient (molar extinction coefficient). For one molar solution and when the path length is 1cm then $A = \epsilon$

Thus molar absorptivity coefficient is defined as the absorbance of a molar solution of path length 1cm.

The ratio I/I_0 is Transmittance T Then $A = 1$

Plot of A vs C is a straight line passing through origin.



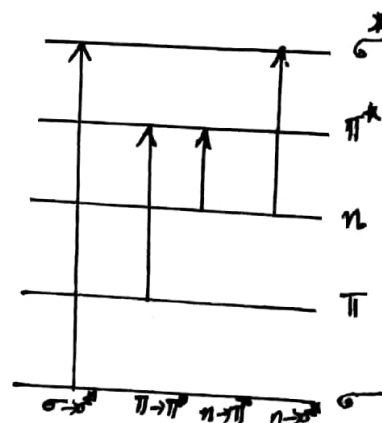
By measuring the absorbance of a sample,

we can determine the concentration of the unknown species

Molecular orbitals and different types of Electronic Transitions

On the basis of Molecular Orbital Theory

(MOT), electrons in polyatomic molecules



can be classified into σ , π or n electrons in bonding molecular orbitals (BMO) and antibonding molecular orbitals (ABMO) are σ^* , π^* .

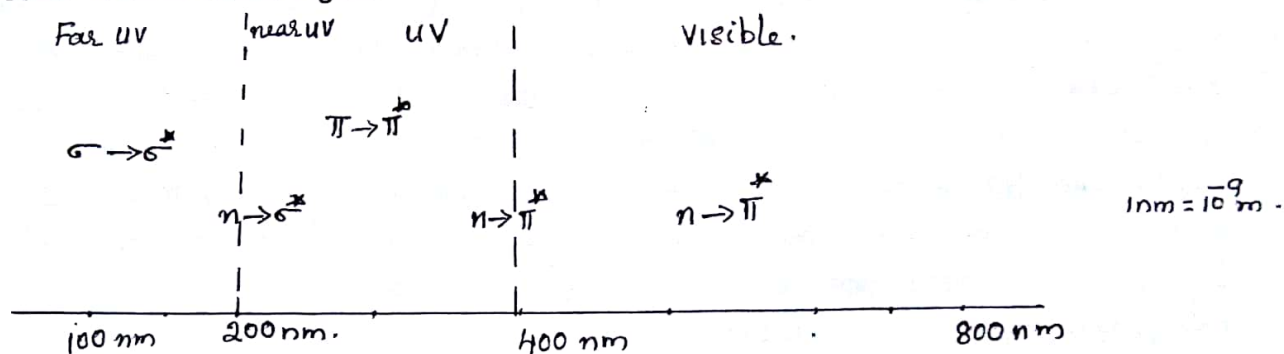
Electronic transitions of the type $\sigma \rightarrow \sigma^*$, $\pi \rightarrow \pi^*$, $n \rightarrow \pi^*$, $n \rightarrow \sigma^*$ are allowed.

$\sigma \rightarrow \sigma^*$ transition :- occur in saturated hydrocarbons and other single bonded compounds. As σ electrons are held more firmly in the molecule, this involve high energy and found in far UV region. (122nm for CH_4 and 135nm for C_2H_6)

$\pi \rightarrow \pi^*$ transition:- This transition takes place in unsaturated compounds, occur in the UV and Visible region.

$n \rightarrow \sigma^*$ occur in near UV region

$n \rightarrow \pi^*$ occur in UV or Visible region.



For aldehydes and ketones, intense band near 180 nm is due to $\pi \rightarrow \pi^*$ transition and the weaker band near 285nm is due to $n \rightarrow \pi^*$ transition.

$n \rightarrow \pi^*$ transition in 250 -360nm indicates the presence of ~~chromophore~~ ^{chromophore} groups such as - C=O, - C=N, - N=N-, NO_2 , -COOH, -COOR, -CONH₂.

$n \rightarrow \sigma^*$ transition at 220nm indicates the presence of amines, ethers, alcohols etc.

Electronic spectroscopy is particularly very useful to study conjugated and aromatic systems.

$\pi \rightarrow \pi^*$ transition at 217nm is due to conjugated diene and is considered as 'parent' to whole series of conjugated dienes.

Shifts in absorption bands and their intensity changes are referred to as follows.

Bathochromic Shift – red shift - A shift of λ_{max} to longer wave length region.

Hysochromic shift – blue shift - A shift to shorter wave length.

Hyperchromic shift - increase in intensity of absorption band.

Hypochromic shift - decrease in intensity of absorption band.

Applications of Electronic spectra

1. For characterization of aromatic compounds and conjugated dienes and other unsaturated compounds.
2. Determination of molecular weights of unknown compounds.
3. Determination of unknown concentration.
4. Study of Kinetics of Chemical reactions.
5. Detection of impurities.
6. Control of purification.

PHOTOELECTRON SPECTROSCOPY (PES)

Photoelectron spectroscopy is an experimental technique for the determination of atomic and molecular energy levels.

When a photon of energy $h\nu$ is incident on an atom or molecule, its energy is transferred to the electrons. Part of this energy is used to overcome the binding energy and the electron is ejected out of the atom or molecule with the remaining energy of photon in the form of $KE = \frac{1}{2}mv^2$.

$$\text{Thus } E = h\nu = \frac{1}{2}mv^2 + BE \dots (1)$$

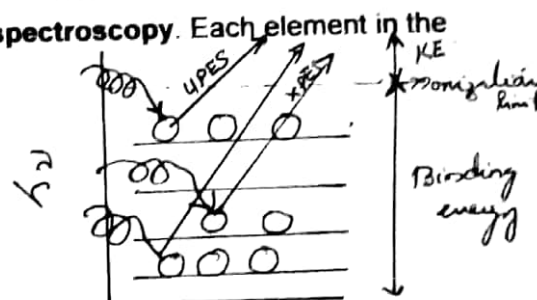
These ejected electrons with different kinetic energies are passed through a hemispherical analyzer and separated according to their binding energies. Each peak in the spectrum corresponds to the ionization of a single atomic orbital. The area under the peak is a measure of degeneracy of that level. In the case of molecules, the number of bands are equal to the number occupied molecular orbitals. Thus PES is a record of core shells and valence shell in atom or molecule.

PES is of two types. (1) Ultraviolet photoelectron spectroscopy (UPES). (2) X-ray photoelectron spectroscopy (XPES).

Ultraviolet photoelectron spectroscopy (UPES): It is used to study the ionization energies of valance shell electrons. A Helium lamp is commonly used as UV source. Wavelength less than 200nm is used because several electron volt is required to eject a valance electron.

X-ray photoelectron spectroscopy (XPES): This is used when core electrons are being studied. Photons of high energies are needed to expel a core electrons. X-rays are used for it. XPES gives lines characteristic of elements present in a compound or alloy. Core ionization energies are characteristic of individual atoms. Hence this technique is also called as **core electron spectroscopy**. Each element in the sample gives a peak in the spectrum as given below

Element	Li	Be	B	C	N	O	P
Kshell ionization energy ev	50	110	190	280	400	530	690



Detection of one of these values indicates the presence of corresponding element. This application is hence called **Electron spectroscopy for chemical analysis (ESCA)**. The exact binding energy of an electron depends on 1. Oxidation state of an atom and 2. The physical environment. Changes in these factors give rise to a small shift in position of peak in spectrum. These changes are called **Chemical shifts**. Higher oxidation states have higher binding energies. For ex: Peak at 455 eV in Ti metal has shifted to 459 eV in TiO_2 .

APPLICATIONS: XPES is very useful for studying

1. Surface state of heterogeneous catalysts.
2. The difference between the surface and bulk structures.
3. The process that can cause damage to high temperature superconductors and semiconductor wafers.
4. The presence of inequivalent atoms of same molecule.
5. The atomic and molecular energy levels, binding energies and ionization energies.

NUCLEAR MAGNETIC RESONANCE SPECTROSCOPY (NMR)

Radiofrequency waves induce transitions between magnetic energy levels of nuclei in presence of an external magnetic field.

In the nucleus of an atom, the proton and neutron each has spin $\frac{1}{2}$. Nuclear spin is the sum of spins of protons and neutrons present in it. The nuclear spin is given by the spin quantum number I .

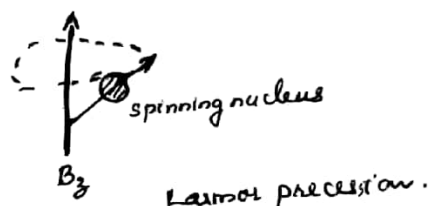
	Mass number	Atomic number	Nuclear spin I	Examples
1.	odd	odd	$I = \text{half integral} - \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \dots$	$^1\text{H}_1, ^{15}\text{N}_7, ^{19}\text{F}_9, ^{31}\text{P}_{15}, \dots$
2.	Even	Even	$I = 0$	$^4\text{He}_2, ^{12}\text{C}_6, ^{16}\text{O}_8, \dots$
3.	Even	odd	$I = \text{integral} - 1, 2, 3, \dots$	$^2\text{H}_1, ^{14}\text{N}_7, ^{10}\text{B}_5, \dots$

The spinning nucleus behave like a tiny bar magnet with a magnetic moment μ along the axis of spin.

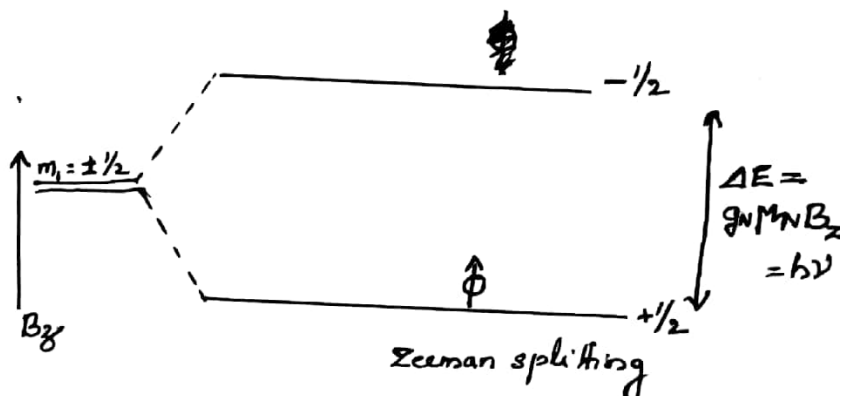
The magnitude of nuclear magnetic moment $\mu = g_N \mu_N \sqrt{I(I+1)}$. Where g_N = Nuclear g factor,

μ_N = Nuclear magneton. The spinning nucleus has $m_I = (2I+1)$ energy states. These components are normally degenerate i.e. all having same energy.

When this spinning nucleus is placed in an external magnetic field B_z in z direction, the nucleus will circulate about the axis of the applied field. This motion is called **Larmor precession**.



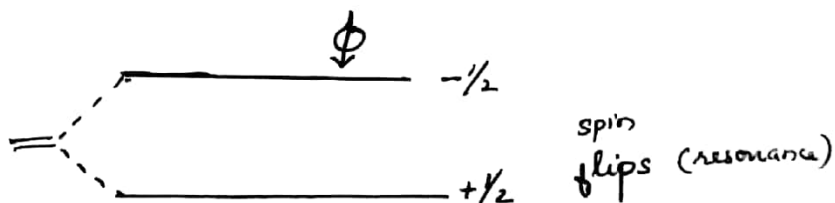
This is due to the interaction of magnetic field of nucleus with the external magnetic field. The degeneracy of the energy levels is removed and $(2I+1)$ different energy levels are formed. This is called **Zeeman splitting** of energy levels. For example a proton ^1H has $I = \frac{1}{2}$, $m_I = \pm \frac{1}{2}$.



The energy difference between the two states is given by $\Delta E = E_{(-1/2)} - E_{(+1/2)} = g_N \mu_N B_z$.

ΔE is directly proportional to the strength of the applied external magnetic field B_z .

This energy difference is equal to the energy of radio frequency wave. If a beam of radio frequency wave is applied on the nucleus placed in the magnetic field, energy



can be exchanged. The nuclear spin "flips" from the upward direction ($\uparrow$) to the downward direction ($\downarrow$), i.e. nucleus jumps to the higher spin state. $\Delta E = g_N \mu_N B_z = h\nu$. A signal peak is obtained when flipping takes place. Then the nucleus is said to be in resonance called **nuclear magnetic resonance**.

Chemical shift: The applied external magnetic field affects the motion of the electrons surrounding the nucleus. The electron cloud tends to circulate. This is called **diamagnetic circulation**. This diamagnetic

field generated by the motion of electrons may oppose or align with the external field. The difference between the magnitudes of the magnetic field at which free nuclei and molecular nuclei resonates is called **chemical shift δ** .

The total field experienced by the nucleus is given by

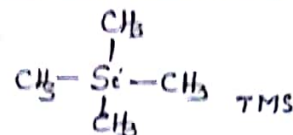
$$B_{\text{effective}} = B_{\text{applied}} - B_{\text{induced}}$$

$$\text{Chemical shift } \delta = \frac{B_{\text{sample}} - B_{\text{ref}}}{B_0} \times 10^6 \text{ ppm.}$$

Chemical shifts are measured by using a reference compound that gives a single sharp resonance peak. Tetramethylsilane (TMS) $(\text{CH}_3)_4\text{Si}$ is used as the reference.

The reasons for using TMS as reference are

1. It is chemically inert.
2. It gives a single sharp peak due to 12 equivalent protons in the same environment.
3. The TMS peak is at highest field strength than all other groups.
4. It has low boiling point and hence easily recovered after recording the spectrum.



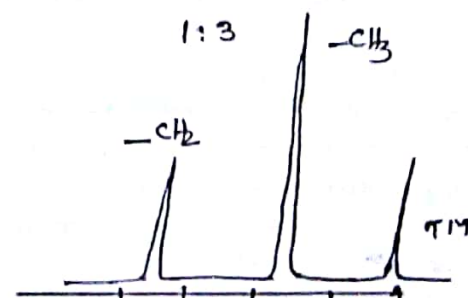
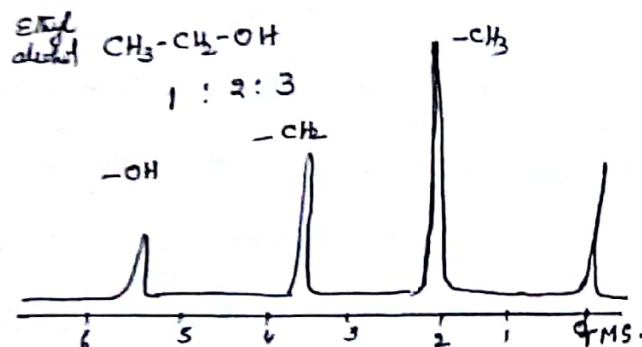
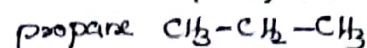
Two Chemical shift scales are used 1. The delta δ scale – by which TMS signal peak is set equal to zero.

2. The Tesla τ scale in which TMS signal peak is set equal to 10. The relation between two is $\delta = 10 - \tau$.

In NMR spectrum of ethyl alcohol three peaks are obtained corresponding to $-\text{OH}$ proton, two $-\text{CH}_2$ protons and three $-\text{CH}_3$ protons. This is because of the different chemical environment of protons. This change in position of the signals of protons due to change in chemical environment is called **chemical shift**.

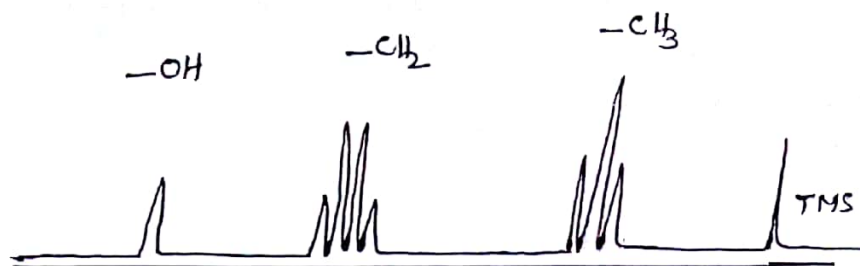
The area under each peak (intensity of peak) is equal to the number of protons in the same environment. The ratio of intensities of peaks is same as the number of protons in different environments. In ethyl alcohol this ratio is 1:2:3 which is same as H atoms in $-\text{OH}$, $-\text{CH}_2$, $-\text{CH}_3$ groups.

In the case of propane $\text{CH}_3-\text{CH}_2-\text{CH}_3$ two end $-\text{CH}_3$ groups have same environment and hence they are equivalent. They will resonate in same frequency while $-\text{CH}_2$ group resonates in another frequency. The ratio intensities will be 2:6 or 1:3.



Spin-spin splitting: Under high resolution, splitting of spectral lines occurs due to inter nuclear spin-spin interactions. The magnetic lines of force of one nucleus interact with the lines of force of neighbouring nuclei. This interaction is called **spin-spin coupling** results in splitting of lines in the spectrum. The resonance peak of one group split into $(n+1)$ peaks, where 'n' is the number of protons in the neighbouring C-atom. Thus High resolution NMR spectrum of acidified ethyl alcohol consists of $-\text{OH}$ singlet, a CH_2 quartet (intensity ratio = 1 : 3 : 3 : 1), and a $-\text{CH}_3$ triplet (1:2:1).

Ethyl alcohol $\text{CH}_3\text{-CH}_2\text{-OH}$.



spin-spin splitting in ethyl alcohol.

Applications of NMR spectroscopy

1. Different types of protons in a molecule can be identified from number of peaks in the spectrum.
2. Splitting of peaks gives the number of protons in the neighbouring C-atom.
3. Number of each kind of protons can be identified from the area under each peak.
4. NMR is widely used in medical field known as MRI scans (Magnetic resonance imaging) used to identify the cancerous cells.
5. To determine the structure of organic molecules.
6. It is used for surface analysis.

SOLID SURFACE CHARACTERISATION

The higher concentration of any molecular species on the surface of a solid than in the bulk is known as **adsorption**.

The solid on which adsorption takes place is called **adsorbent** and substance which is held on the surface is called **adsorbate**.

Physisorption occurs at the surface through van der Waals forces while **chemisorption** takes place through chemical interaction between adsorbent and adsorbate molecules.

Adsorption of Gases by solids: Thermodynamics of adsorption

The adsorption of gases by solids can be studied by measuring the pressure of the gas using a manometer. When adsorbent (solid) is introduced into the vessel containing the gas (adsorbate), the pressure decreases. This is noted on the manometer. This fall in pressure is used to calculate the quantity of gas adsorbed by the solid using Boyle's law.

As a result of adsorption, there is a decrease in residual forces on the surface of adsorbent. There is a decrease in surface energy which appears as heat. Adsorption is accompanied by evolution of heat, i.e. it is an exothermic process. The amount of heat evolved when one mole of a gas is adsorbed on a solid is known as the **molar enthalpy of adsorption**. ΔH is $-ve$.

Since adsorbate (gas) changes from random gaseous state to the less random adsorbed state on the surface of solid, entropy decreases. ΔS is $-ve$.

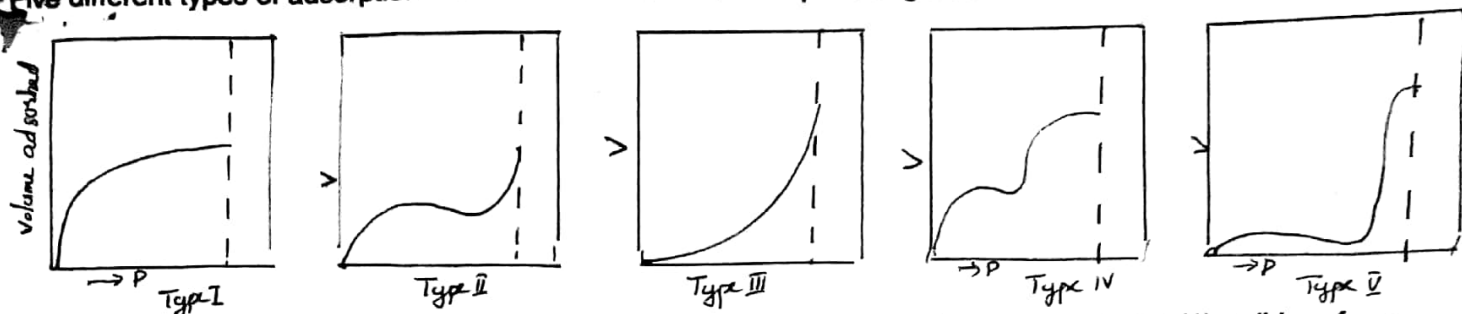
For a spontaneous process the free energy change $\Delta G = \Delta H - T\Delta S$ should be $-ve$. Since ΔH and $T\Delta S$ are both negative, it is evident that ΔH should have a high negative value for ΔG to be negative. But by experiment it is found that the heat of adsorption per mole of adsorbate goes on decreasing and finally $\Delta H = T\Delta S$ and $\Delta G = 0$. At this stage equilibrium is attained.

Factors affecting adsorption: 1. Temperature 2. Pressure 3. Nature of adsorbate 4. Nature of adsorbent.

Adsorption Isotherm: Graph showing variation of adsorption with pressure at constant temperature.

Adsorption Isobar: Graph showing the effect of temperature on the extent of adsorption at constant pressure.

Adsorption Isostere: Graph showing variation of pressure with temperature for a given amount of adsorption.



The BET theory of multilayer adsorption of gases: In BET theory, it is assumed that (1) solid surface possesses uniform, localized sites. Adsorption at one site does not affect adsorption at neighbouring site. (2) Molecules can be adsorbed in multilayers, 1, 2, upto nth layers (3) The energy of adsorption in the first layer E_1 is constant and energy of adsorption of other layers is assumed to be the same as E_L , the energy of liquefaction of the gas.

Based on the above assumptions, Brunauer, Emmett and Teller derived the following equation known as BET equation.

$$\frac{P}{V_{\text{total}}(P_0 - P)} = \frac{1}{V_{\text{mono}} C} + \frac{C - 1}{V_{\text{mono}} C} \left[\frac{P}{P_0} \right]$$

Where, V_{total} = The volume of gas adsorbed at pressure P .

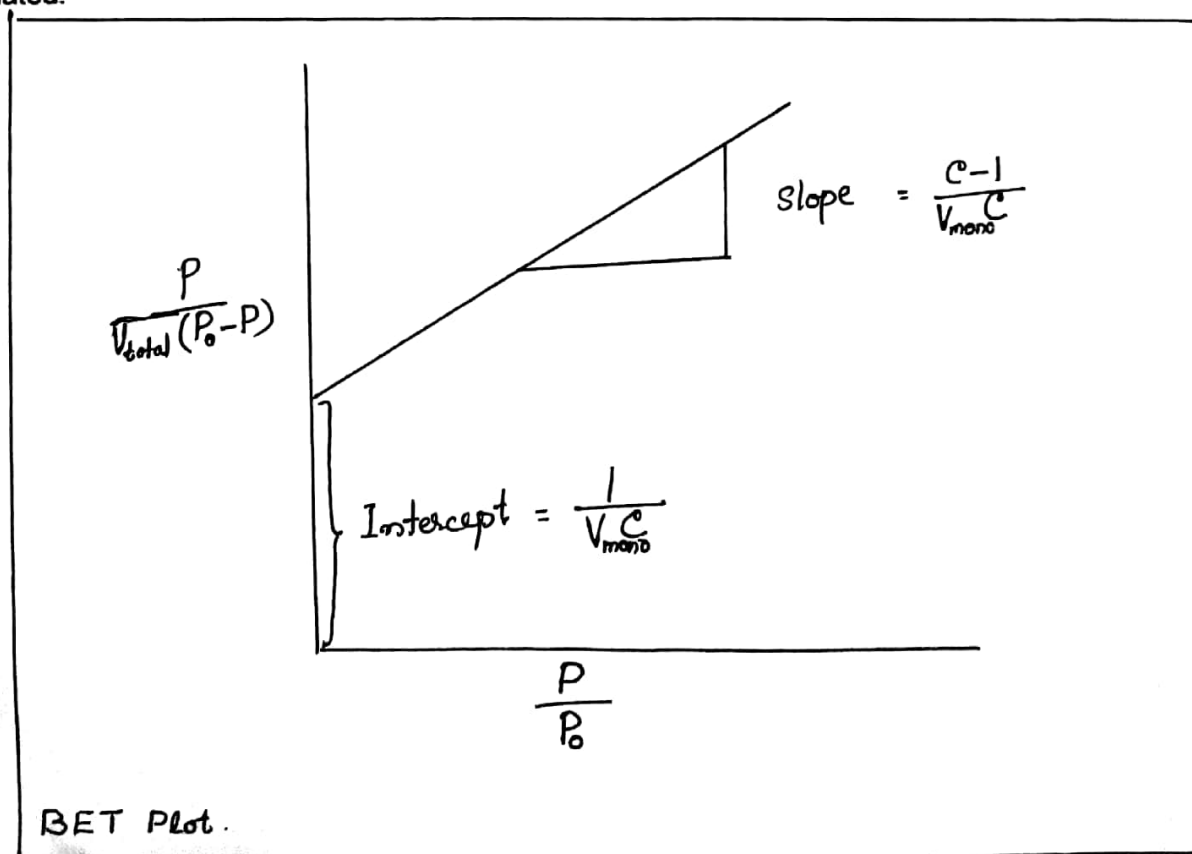
V_{mono} = The volume of gas adsorbed when the surface of solid is covered completely with mono layer of the adsorbed gas.

C = a constant depending upon the nature of gas, $C = e^{(E_1 - E_L)/RT}$

P_0 = pressure at the liquefaction of the gas

BET equation is of the form $Y = mx + C$. A plot of $\frac{P}{V_{\text{total}}(P_0 - P)}$ against $\frac{P}{P_0}$ should give a straight line.

The slope of linear plot gives $\frac{C - 1}{V_{\text{mono}} C}$, Intercept gives $\frac{1}{V_{\text{mono}} C}$ from which both V_{mono} and C can be Calculated.





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From V_{mono} (1) the surface area of the adsorbent can be calculated (2) the area of cross section of the molecule can be determined from the density of the liquefied adsorbate and from the molar mass of the adsorbate. (3) The volume of single molecule can also be calculated as follows.

Density $\rho = \frac{M}{NV}$ Where M =molar mass, N =Avogadro's number, V =Volume of single molecule

Or $V = \frac{M}{\rho N}$

Assuming the molecule to be spherical with radius r , $V = \frac{4}{3}\pi r^3 = \frac{M}{\rho N}$

$$\therefore r = \left[\frac{3M}{4\pi\rho N} \right]^{1/3}$$

$$\therefore \text{area of cross section 'a' of a single molecule of adsorbate} = \pi r^2 = \pi \left[\frac{3M}{4\pi\rho N} \right]^{2/3}$$

$$\therefore \text{surface area of adsorbent } s = \frac{V_a N w}{22400} \quad (w = \text{weight of solid adsorbent})$$